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PREDICTIVE ANALYTICS OVERVIEW

Customer Behaviour and Sentiment Analysis at Turtle Games

LSE DATA ANALYTICS CAREER ACCELERATOR

Executive Summary

Turtle Games wanted to gain a better understanding of customer behaviour to improve loyalty, marketing and customer satisfaction. This report combines Natural Language Processing (NLP), multiple linear regression, decision tree regression and K-means clustering to identify the main factors influencing customer loyalty and customer opinions.

The findings show that customer reviews are generally positive, with customers valuing enjoyable gameplay and educational products. Spending score was identified as the strongest predictor of loyalty points, while decision tree analysis confirmed its importance. Customer segmentation also identified five distinct customer groups with different purchasing behaviours. Overall, the results suggest that Turtle Games should focus on behaviour-based marketing, improve product quality and continue using customer feedback to support business decisions.

1. Introduction

Turtle Games aims to better understand its customers by analysing purchasing behaviour and online reviews. The objectives of this project were to identify the factors influencing customer loyalty, group customers into meaningful market segments and analyse customer opinions to support business decision making.

To achieve these objectives, Natural Language Processing, multiple linear regression, decision tree regression and K-means clustering were applied. Using several analytical techniques provides a broader understanding of customer behaviour than relying on a single method.

2. Dataset and Methodology

The analysis was carried out using a dataset containing 2,000 customer records. The dataset included demographic information, behavioural variables such as spending score, remuneration and loyalty points, together with customer reviews.

Before modelling, the data were cleaned and checked for missing values. Customer reviews were converted to lowercase, punctuation and duplicate reviews were removed, and common stop words were excluded before conducting the NLP analysis.

Exploratory data analysis was first carried out in R to understand the distribution of the data. Multiple linear regression was then used to identify the variables that best predict loyalty points. Decision tree regression was applied to provide an interpretable prediction model, while K-means clustering identified customer segments based on spending behaviour.

3. Customer Review Analysis

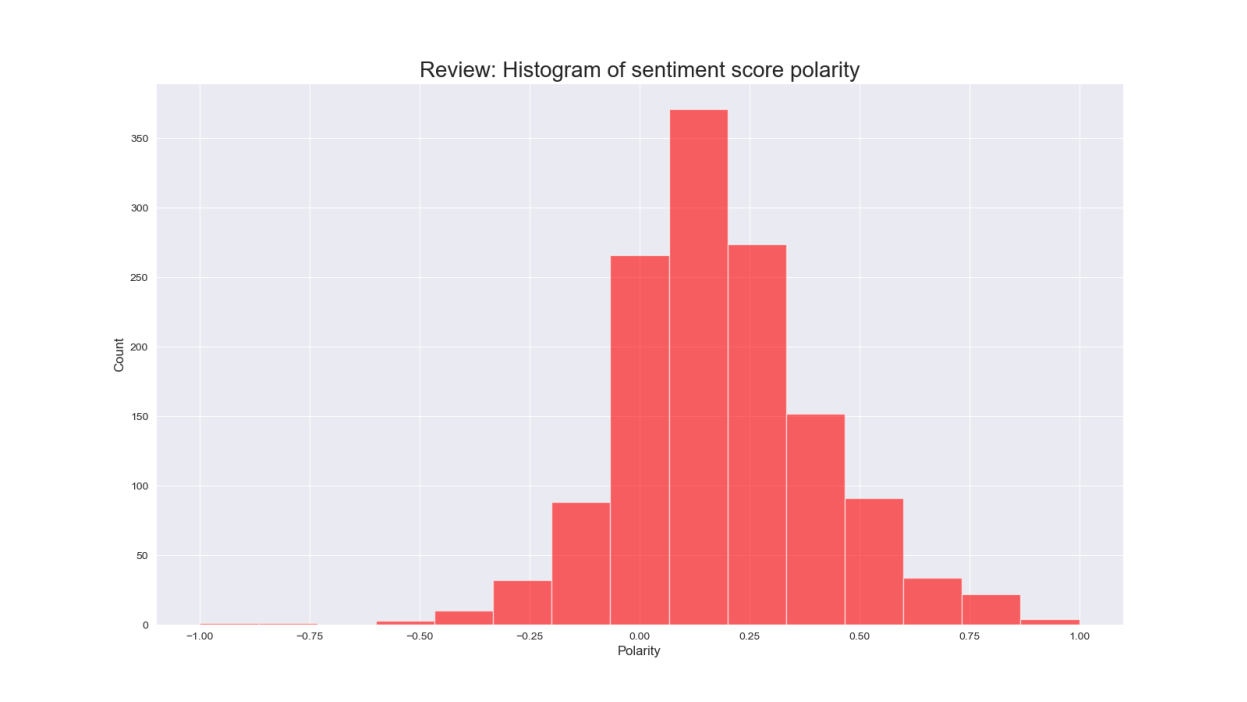


Figure 1. Review Sentiment Histogram

Natural Language Processing was used to understand customer opinions by analysing review sentiment and frequently used words. The sentiment analysis showed that customer feedback is generally positive, with most reviews expressing favourable opinions of Turtle Games' products. Only a relatively small number of reviews showed strong negative sentiment.


game
turtle
play
one
card
great
book
make
time
use
piece
new
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playing
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old
recommend
good
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actually
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adult
thought
others
lords waterdeep
game play
overall
daughter
child
something
size
yarn
isnt
didnt
cardboard
animal
acquire
youll
right
able
dog
gets
edition
hold
difficult
put
magnet
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doesnt
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come
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part
question
kit
give
quest
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great game
far
open
son
playing
family
people
friend
ball whacks
year
may
old
recommend
feel
never
best
number
building
day
instruction
idea
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long
easily
going
everything
ball
money
year
two
cute
board
anger
three
lords waterdeep
toy
product
strategy
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open
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great game
quest
give
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kid
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toy
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The most common words included *game*, *play*, *fun*, *great*, *cards* and *tiles*, suggesting that customers mainly discuss gameplay and product quality. Positive reviews frequently mentioned enjoyable gameplay, educational value and products being suitable as gifts.

Negative reviews mainly focused on poor product quality, incomplete components and confusing instructions rather than the games themselves. This suggests that customer dissatisfaction is largely related to manufacturing or packaging issues rather than the overall gaming experience.

The NLP (Natural Language Processing) analysis provides valuable business insight by identifying both the strengths of Turtle Games' products and areas where improvements could increase customer satisfaction.

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4. Predicting Customer Loyalty

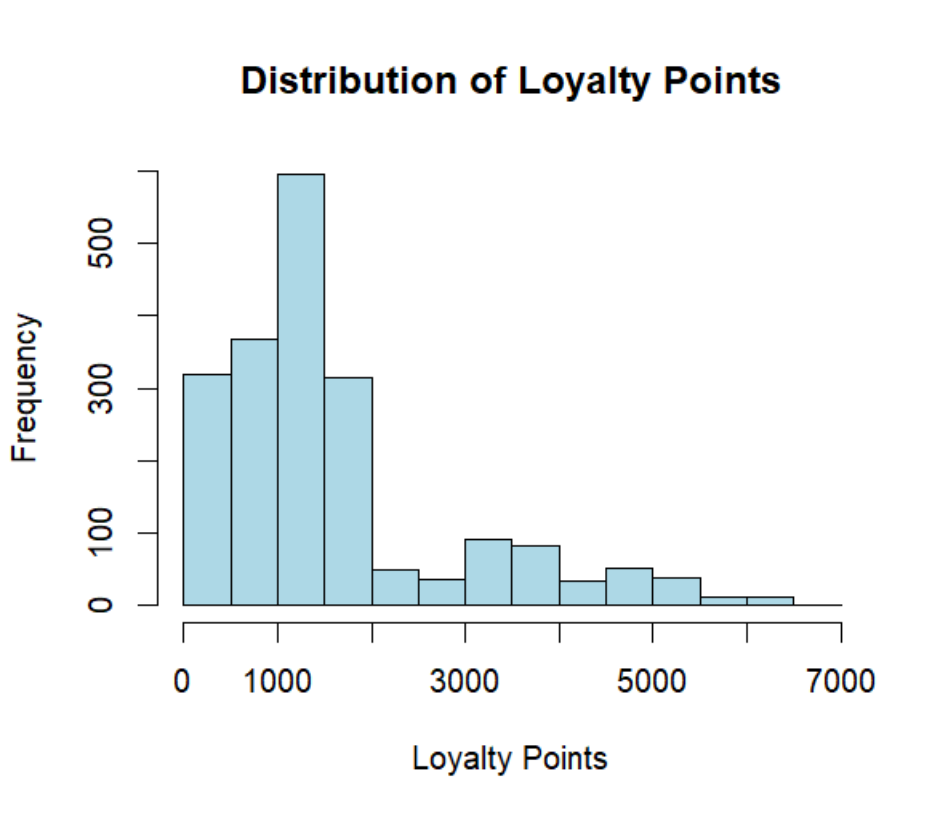


Figure 3. Distribution of Loyalty Points

Exploratory data analysis showed that most customers had relatively low to moderate loyalty point balances, while only a small number accumulated very high balances. This positive skew suggests that the loyalty programme is driven by a relatively small group of highly loyal

customers.Correlation analysis found that spending score and remuneration had the strongest positive relationships with loyalty points, while age showed only a weak relationship. A multiple linear regression model was then developed using age, remuneration, spending score and product. Product was not statistically significant and was removed from the final model.

The simplified model explained approximately **84%** of the variation in loyalty points, indicating that customer behaviour is a strong predictor of loyalty. Spending score was identified as the most influential variable, followed by remuneration, while age contributed relatively little to the model.

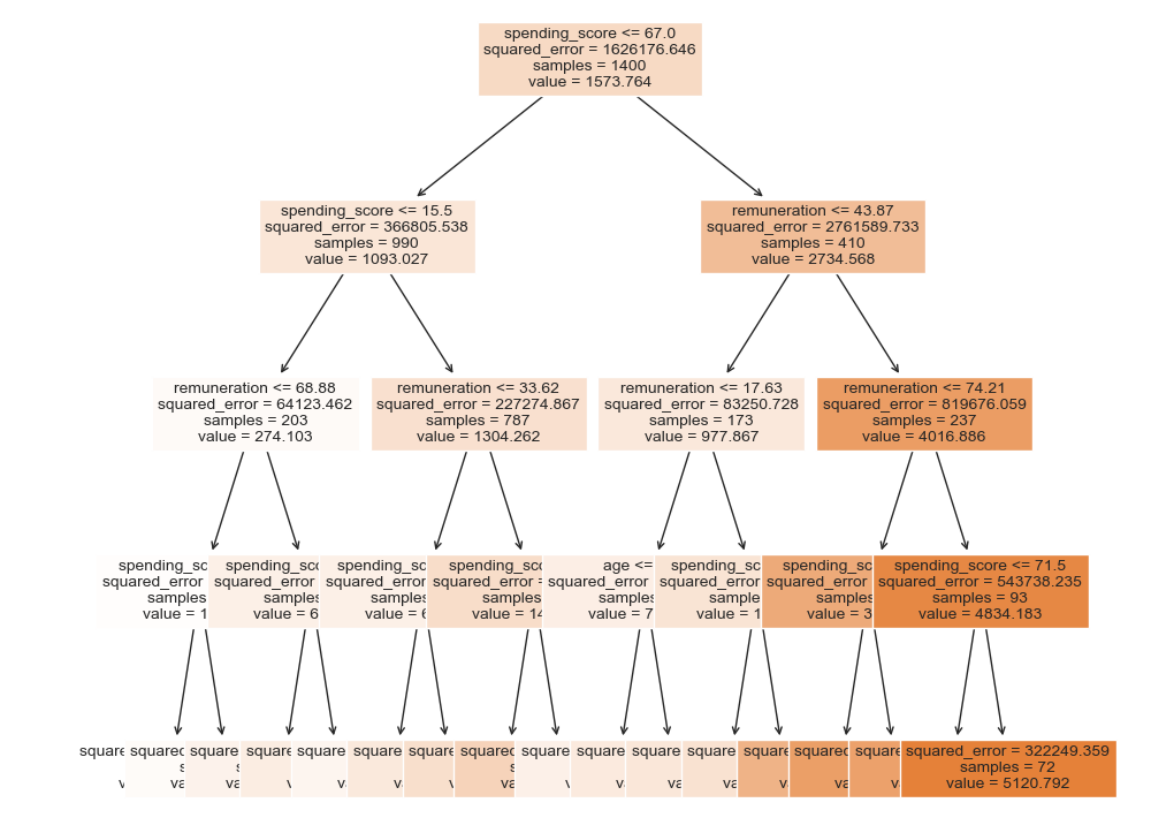


Figure 4. Pruned Decision Tree

The decision tree analysis supported the regression results. Spending score formed the first split within the tree, showing that it is the most important factor influencing customer loyalty. Remuneration provided additional predictive power, while age had a much smaller impact. The decision tree also provides an easy-to-understand model that can be used by non-technical stakeholders when identifying high-value customers.



Figure 5. Customer Segments

K-means clustering identified five customer segments with different purchasing behaviours. The analysis showed that customers with similar incomes often have different spending patterns, demonstrating that behavioural segmentation provides more useful business insight than demographic information alone. These customer groups can be used to develop more personalised marketing campaigns and improve customer retention.

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5. Recommendations

The findings suggest that Turtle Games should focus its marketing activities on customer behaviour rather than demographic characteristics. Customers with high spending scores should receive personalised rewards and loyalty incentives to encourage long-term retention.

The customer segments identified through K-means clustering provide an opportunity to tailor marketing campaigns according to purchasing behaviour instead of treating all customers the same. At the same time, customer reviews should continue to be monitored to identify recurring issues and measure customer satisfaction over time.

Finally, addressing common complaints about product quality and confusing instructions is likely to improve customer satisfaction and reduce negative reviews while maintaining the positive aspects of gameplay that customers already value.

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6. Conclusion

This analysis demonstrated how combining Natural Language Processing, multiple linear regression, decision tree regression and customer segmentation can provide valuable business insight. Spending behaviour was consistently identified as the strongest predictor of customer loyalty, while customer reviews highlighted both the strengths of Turtle Games' products and opportunities for improvement.

By focusing on behaviour-based marketing, improving product quality and continuing to analyse customer feedback, Turtle Games can strengthen customer loyalty, improve customer satisfaction and support future business growth.